That is $S = \{TT, TH, HT, HH\}$

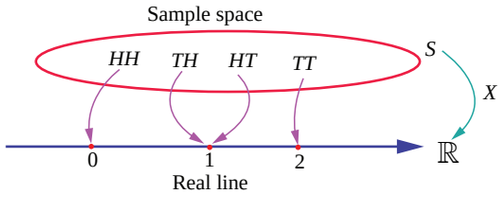
(ii) Let $X : S \to \mathbb{R}$ be the number of tails

Then $X(TT) = 2$ (2 Tails)

$X(TH) = 1$ (1 Tail)

$X(HT) = 1$ (1 Tail)

and $X(HH) = 0$ (0 Tails).



A mapping X (.) from S to $\mathbb{R}$

Fig. 11.2

Then X is a random variable that takes on the values 0, 1 and 2.

Let $X(\omega)$ denote the number of tails, this gives

$$X(\omega) = \begin{cases} 2 & \text{if } \omega = TT \\ 1 & \text{if } \omega = HT, TH \\ 0 & \text{if } \omega = HH \end{cases}$$

The inverse images of 1 is $\{TH, HT\}$. That is $X^{-1}(\{1\}) = \{TH, HT\}$.

(iii) Number of elements in inverse images are shown in the table.

Values of the Random Variable	0	1	2	Total
Number of elements in inverse image	1	2	1	4

Example 11.2

Suppose a pair of unbiased dice is rolled once. If X denotes the total score of two dice, write down (i) the sample space (ii) the values taken by the random variable X, (iii) the inverse image of 10, and (iv) the number of elements in inverse image of X.

Solution

(i) The sample space

S = $\{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}$,

consists of 36 ordered pairs (α, β) where α and β can take any integer value between 1 and 6 as shown. X is assigned to each point (α, β) the sum of the numbers on the dice .

$$S = \begin{Bmatrix} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{Bmatrix}$$

That is $X(\alpha, \beta) = \alpha + \beta$.

Therefore

$X(1,1) = 1+1 = 2$

$X(1,2) = X(2,1) = 3$

$X(1,3) = X(2,2) = X(3,1) = 4$

$X(1,4) = X(2,3) = X(3,2) = X(4,1) = 5$

$X(1,5) = X(2,4) = X(3,3) = X(4,2) = X(5,1) = 6$

$X(1,6) = X(2,5) = X(3,4) = X(4,3) = X(5,2) = X(6,1) = 7$

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 181 16-12-2022 16:50:09